Chapter 1. Measure Spaces, Measurable Functions and Integration

σ-algebras, Dynkin systems, Lebesgue measure, measurable functions, the Lebesgue integral and its limit theorems, L^p-spaces

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1.1 Systems of sets and measures

Motivation. Measures live on families closed under countable operations, and are determined by any $\cap$-stable generator.

Definition 1.1 (σ-algebra)

Let Ω be a set and $\mathcal{P}(\Omega) := \{A \mid A \subset \Omega\}$ its power set. A family $\mathcal{A} \subset \mathcal{P}(\Omega)$ is a **σ-algebra** (or σ-field) on Ω if

(i) $\Omega \in \mathcal{A}$;

(ii) $A \in \mathcal{A} \Rightarrow A^c := \Omega \setminus A \in \mathcal{A}$;

(iii) $A_i \in \mathcal{A}$, $i \in \mathbb{N} \Rightarrow \bigcup_{i\in\mathbb{N}} A_i \in \mathcal{A}$.

"σ" signals *countable* operations.

Example 1.2

(i) $\mathcal{P}(\Omega)$ is a σ-algebra (natural for countable Ω, otherwise too large).

(ii) If $\mathcal{A}_i$, $i \in I$ (I arbitrary, possibly uncountable), are σ-algebras on Ω, then $\bigcap_{i\in I} \mathcal{A}_i = \{A \subset \Omega \mid A \in \mathcal{A}_i \ \forall i\}$ is a σ-algebra.

(iii) (*Generated σ-algebra*) For $\emptyset \neq \mathcal{A}_0 \subset \mathcal{P}(\Omega)$,

$$\sigma(\mathcal{A}_0) := \bigcap_{\mathcal{B}\ \sigma\text{-algebra},\ \mathcal{A}_0\subset\mathcal{B}} \mathcal{B}$$

is a σ-algebra by (ii) and is the smallest σ-algebra containing $\mathcal{A}_0$.

(iv) (*Borel sets*) If Ω is a topological space and $\mathcal{A}_0$ the family of open sets, $\mathcal{B}(\Omega) := \sigma(\mathcal{A}_0)$ is the **Borel σ-algebra**. Open sets, closed sets and countable unions of closed sets are Borel; nevertheless $\mathcal{B}(\mathbb{R}) \neq \mathcal{P}(\mathbb{R})$.

(v) (*Trace*) For a σ-algebra $\mathcal{A}$ on Ω and $\Omega' \subset \Omega$, $\Omega' \cap \mathcal{A} := \{A \cap \Omega' \mid A \in \mathcal{A}\}$ is a σ-algebra on Ω'. If $\Omega' \in \mathcal{A}$, then $\Omega' \cap \mathcal{A} = \{A \in \mathcal{A} \mid A \subset \Omega'\}$.

Definition 1.3 (Measure)

Let $\Omega \neq \emptyset$ and $\mathcal{A}$ a σ-algebra on Ω. A map $\mu : \mathcal{A} \to [0, \infty]$ is a **measure** on $(\Omega, \mathcal{A})$ if

(i) $\mu(\emptyset) = 0$;

(ii) (σ-additivity) $\mu\big(\dot{\bigcup}_{i\in\mathbb{N}} A_i\big) = \sum_{i=1}^{\infty} \mu(A_i)$ for pairwise disjoint $A_i \in \mathcal{A}$.

$(\Omega, \mathcal{A}, \mu)$ is a **measure space**, $(\Omega, \mathcal{A})$ a **measurable space**, and the elements of $\mathcal{A}$ are **measurable sets**. If moreover $\mu(\Omega) = 1$, μ is a **probability measure**; we then write $P := \mu$, call $(\Omega, \mathcal{A}, P)$ a **probability space** and its measurable sets **events**.

Notation. $\dot{\bigcup}_{i\in\mathbb{N}} A_i$ (or $A_1 \,\dot\cup\, A_2$) denotes a union whose members are pairwise disjoint.

Example 1.4

(i) (*Dirac measure*) For $x \in \mathbb{R}^d$, $\delta_x(A) := 1$ if $x \in A$ and 0 otherwise defines a probability measure on $(\mathbb{R}^d, \mathcal{B}(\mathbb{R}^d))$.

(ii) (*Restriction*) If $(\Omega, \mathcal{A}, \mu)$ is a measure space and $\Omega' \in \mathcal{A}$, then $(\Omega', \mathcal{A} \cap \Omega', \mu)$ is again a measure space.

(iii) (*Counting measure*) On $(\mathbb{N}_0, \mathcal{P}(\mathbb{N}_0))$, $\mu(A) := |A| \in [0, \infty]$ (number of elements) is a measure.

Dynkin systems (λ-systems) are easier to verify and serve uniqueness arguments.

Definition 1.5 (Dynkin system)

$\mathcal{D} \subset \mathcal{P}(\Omega)$ is a **Dynkin system** if (i) $\Omega \in \mathcal{D}$;
(ii) $A \in \mathcal{D} \Rightarrow A^c \in \mathcal{D}$;
(iii) $A_i \in \mathcal{D}$, $i \in \mathbb{N}$, pairwise disjoint $\Rightarrow \dot{\bigcup}_i A_i \in \mathcal{D}$.

Unlike in Definition 1.1, condition (iii) is for disjoint sequences only.

Example 1.6

(i) Every σ-algebra is a Dynkin system.

(ii) If P_1, P_2 are probability measures on $(\Omega, \mathcal{A})$, then $\mathcal{D} := \{A \in \mathcal{A} \mid P_1(A) = P_2(A)\}$ is a Dynkin system (total mass 1, $P_i(A^c) = 1 - P_i(A)$, σ-additivity), in general *not* a σ-algebra.

Remark 1.7

(i) If $\mathcal{D}$ is a Dynkin system and $A, B \in \mathcal{D}$ with $A \subset B$, then $B \setminus A = (B^c \,\dot\cup\, A)^c \in \mathcal{D}$.

(ii) A Dynkin system that is $\cap$-stable (closed under finite intersections) is a σ-algebra.

(a) For $A, B \in \mathcal{D}$: $A \cup B = A \,\dot\cup\, (B \setminus (A \cap B)) \in \mathcal{D}$, by (i) and $\cap$-stability.

(b) For $A_i \in \mathcal{D}$: $\bigcup_i A_i = \dot{\bigcup}_i \left(A_i \cap \left(\bigcup_{n<i} A_n\right)^c\right)$, and each piece lies in $\mathcal{D}$ by (a), $\cap$-stability and closure under complements.

Theorem 1.8 (π–λ theorem)

Let $\mathcal{A}_0 \subset \mathcal{P}(\Omega)$ be $\cap$-stable. Then $\sigma(\mathcal{A}_0) = \mathcal{D}(\mathcal{A}_0)$, where

$$\mathcal{D}(\mathcal{A}_0) := \bigcap_{\mathcal{D} \text{ Dynkin system},\ \mathcal{A}_0 \subset \mathcal{D}} \mathcal{D}$$

is the Dynkin system generated by $\mathcal{A}_0$.

Proof.

Step 1. $\sigma(\mathcal{A}_0)$ is a Dynkin system containing $\mathcal{A}_0$, so $\mathcal{D}(\mathcal{A}_0) \subset \sigma(\mathcal{A}_0)$.

Step 2. For the converse it suffices, by Remark 1.7(ii), to show that $\mathcal{D}(\mathcal{A}_0)$ is $\cap$-stable. For $D \in \mathcal{D}(\mathcal{A}_0)$ put

$$\mathcal{D}_D := \{Q \subset \Omega \mid Q \cap D \in \mathcal{D}(\mathcal{A}_0)\}.$$

It is a Dynkin system: $\Omega \cap D = D$; $Q^c \cap D = D \setminus (Q \cap D)$ (Remark 1.7(i)); disjoint unions pass through $\cap D$.

Step 3. For $E \in \mathcal{A}_0$, $\mathcal{A}_0 \subset \mathcal{D}_E$ ($\cap$-stability), so $\mathcal{D}(\mathcal{A}_0) \subset \mathcal{D}_E$: $E \cap D \in \mathcal{D}(\mathcal{A}_0)$ for all $D \in \mathcal{D}(\mathcal{A}_0)$.

Step 4. So $\mathcal{A}_0 \subset \mathcal{D}_D$, hence $\mathcal{D}(\mathcal{A}_0) \subset \mathcal{D}_D$ for every $D \in \mathcal{D}(\mathcal{A}_0)$: $\cap$-stability. ∎

Theorem 1.9 (Uniqueness of measures)

Let μ_1, μ_2 be measures on $(\Omega, \mathcal{A})$ and $\mathcal{A}_0 \subset \mathcal{A}$ a $\cap$-stable family with $\sigma(\mathcal{A}_0) = \mathcal{A}$ and $\mu_1 = \mu_2$ on $\mathcal{A}_0$. Suppose there are $\Omega_n \in \mathcal{A}_0$ with $\Omega_n \nearrow \Omega$ (i.e. $\Omega_n \subset \Omega_{n+1}$, $\bigcup_n \Omega_n = \Omega$) and $\mu_1(\Omega_n) = \mu_2(\Omega_n) < \infty$. Then $\mu_1 = \mu_2$. If μ_1, μ_2 are probability measures, the sets Ω_n are not needed.

Proof.

Step 1 (probability measures). By Example 1.6(ii), $\mathcal{D} := \{A \in \mathcal{A} \mid \mu_1(A) = \mu_2(A)\}$ is a Dynkin system containing $\mathcal{A}_0$. By Theorem 1.8, $\mathcal{A} = \sigma(\mathcal{A}_0) = \mathcal{D}(\mathcal{A}_0) \subset \mathcal{D}$, so $\mu_1 = \mu_2$.

Step 2 (localisation). For $E \in \mathcal{A}_0$ with $\mu_1(E) = \mu_2(E) < \infty$, $\mathcal{D}_E := \{A \in \mathcal{A} \mid \mu_1(A \cap E) = \mu_2(A \cap E)\}$ is a Dynkin system ($\mu_i(A^c \cap E) = \mu_i(E) - \mu_i(A \cap E)$ by finiteness) containing $\mathcal{A}_0$; Theorem 1.8 gives $\mathcal{D}_E = \mathcal{A}$.

Step 3. $E = \Omega_n$ and continuity from below (Theorem 1.15(ii)): $\mu_1(A) = \lim_n \mu_1(A \cap \Omega_n) = \lim_n \mu_2(A \cap \Omega_n) = \mu_2(A)$ for every $A \in \mathcal{A}$. ∎

Measures are first defined on rings, then extended.

Definition 1.10 (Rings, measures on rings, σ-finiteness)

(i) $\mathcal{R} \subset \mathcal{P}(\Omega)$ is a **ring** if $A, B \in \mathcal{R} \Rightarrow A \setminus B \in \mathcal{R}$ and $A \cup B \in \mathcal{R}$. Then $\emptyset \in \mathcal{R}$ and $A \cap B = A \setminus (A \setminus B) \in \mathcal{R}$, so rings are $\cap$-stable. A ring is a **σ-ring** if it is closed under countable unions; then also $\bigcap_n A_n = A_1 \setminus \bigcup_n (A_1 \setminus A_n) \in \mathcal{R}$. A ring containing Ω is an **algebra**, and a σ-ring containing Ω is a σ-algebra.

(ii) A **measure on a ring** $\mathcal{R}$ is defined as in Definition 1.3, with σ-additivity required only for disjoint $A_i \in \mathcal{R}$ whose union lies in $\mathcal{R}$.

(iii) A measure μ on a ring $\mathcal{R}$ is **σ-finite** if there are $\Omega_n \in \mathcal{R}$ with $\Omega_n \nearrow \Omega$ and $\mu(\Omega_n) < \infty$.

Theorem 1.11 (Carathéodory's extension theorem)

Let μ_0 be a σ-finite measure on a ring $\mathcal{R}$. Then there is a unique σ-finite measure μ on $\sigma(\mathcal{R})$ with $\mu = \mu_0$ on $\mathcal{R}$.

Proof [Supplement].

Step 1 (uniqueness). $\mathcal{R}$ is $\cap$-stable and $\mu_1(\Omega_n) = \mu_2(\Omega_n) = \mu_0(\Omega_n) < \infty$; apply Theorem 1.9.

Step 2 (outer measure). For $Q \subset \Omega$ put

$$\mu^*(Q) := \inf\Big\{ \sum_{n=1}^{\infty} \mu_0(A_n) \;\Big|\; A_n \in \mathcal{R},\ Q \subset \bigcup_n A_n \Big\}.$$

Covers exist ($\Omega_n \nearrow \Omega$); $\mu^*(\emptyset) = 0$, μ^* is monotone and σ-subadditive: given $Q = \bigcup_k Q_k$ with $\mu^*(Q_k) < \infty$ and $\varepsilon > 0$, choose covers $(A_{kn})_n$ of Q_k with $\sum_n \mu_0(A_{kn}) \le \mu^*(Q_k) + \varepsilon 2^{-k}$; then $(A_{kn})_{k,n}$ covers Q and $\mu^*(Q) \le \sum_k \mu^*(Q_k) + \varepsilon$.

Step 3 ($\mu^* = \mu_0$ on $\mathcal{R}$). $\mu^*(A) \le \mu_0(A)$ (cover by A). If $A \subset \bigcup_n A_n$, $A_n \in \mathcal{R}$, the disjoint $B_n := A \cap A_n \setminus \bigcup_{k<n} A_k \in \mathcal{R}$ have union A, so $\mu_0(A) = \sum_n \mu_0(B_n) \le \sum_n \mu_0(A_n)$. Hence $\mu_0(A) \le \mu^*(A)$.

Step 4 (Carathéodory-measurable sets). Let $\mathcal{M}^* := \{A \subset \Omega \mid \mu^*(Q) = \mu^*(Q \cap A) + \mu^*(Q \setminus A)\ \forall Q \subset \Omega\}$. Only "$\ge$" for $\mu^*(Q) < \infty$ needs checking. $\Omega \in \mathcal{M}^*$, and complements by symmetry. For $A, B \in \mathcal{M}^*$,

$$\mu^*(Q) = \mu^*(Q \cap A) + \mu^*(Q \cap A^c \cap B) + \mu^*(Q \cap A^c \cap B^c) \ge \mu^*(Q \cap (A \cup B)) + \mu^*(Q \setminus (A \cup B)),$$

by $Q \cap (A \cup B) = (Q \cap A) \cup (Q \cap A^c \cap B)$: an algebra. For disjoint $A_1, \dots, A_n \in \mathcal{M}^*$, by induction $\mu^*(Q \cap \bigcup_{i \le n} A_i) = \sum_{i \le n} \mu^*(Q \cap A_i)$. Now let $A_i \in \mathcal{M}^*$ be disjoint, $A := \bigcup_i A_i$, $S_n := \bigcup_{i \le n} A_i \in \mathcal{M}^*$. Then

$$\mu^*(Q) = \mu^*(Q \cap S_n) + \mu^*(Q \setminus S_n) \ge \sum_{i \le n} \mu^*(Q \cap A_i) + \mu^*(Q \setminus A),$$

and as $n \to \infty$, by σ-subadditivity, $\mu^*(Q) \ge \sum_i \mu^*(Q \cap A_i) + \mu^*(Q \setminus A) \ge \mu^*(Q \cap A) + \mu^*(Q \setminus A)$. So $A \in \mathcal{M}^*$, and disjointifying (Remark 1.7(ii)(b)) makes $\mathcal{M}^*$ a σ-algebra; $Q = A$ gives σ-additivity.

Step 5 ($\mathcal{R} \subset \mathcal{M}^*$). For $A \in \mathcal{R}$, $\mu^*(Q) < \infty$, and a cover $(A_n) \subset \mathcal{R}$ of Q with $\sum \mu_0(A_n) \le \mu^*(Q) + \varepsilon$: $(A_n \cap A)$ covers $Q \cap A$, $(A_n \setminus A)$ covers $Q \setminus A$, and $\mu_0(A_n) = \mu_0(A_n \cap A) + \mu_0(A_n \setminus A)$, so $\mu^*(Q \cap A) + \mu^*(Q \setminus A) \le \mu^*(Q) + \varepsilon$.

Step 6. $\sigma(\mathcal{R}) \subset \mathcal{M}^*$, and $\mu := \mu^*|_{\sigma(\mathcal{R})}$ extends μ_0 (Step 3), σ-finite as $\mu(\Omega_n) < \infty$. ■

Remark 1.12

The lectures construct Lebesgue measure directly (Section 1.2, after Rudin, *Principles*, Ch. 11): the same outer-measure idea on the ring of elementary sets, using compactness, with measurability defined by approximation and σ-additivity obtained only on the resulting σ-ring.

1.2 Construction of the Lebesgue measure

Motivation. Extend box volume σ-additively: outer measure, a pseudo-metric on sets, then sets approximable by elementary sets.

Definition 1.13 (Additive and σ-additive set functions)

Let $\varphi : \mathcal{R} \to [0, \infty]$ on a ring $\mathcal{R}$ with $\varphi(\emptyset) = 0$. φ is **additive** if

$$A, B \in \mathcal{R},\ A \cap B = \emptyset \implies \varphi(A \cup B) = \varphi(A) + \varphi(B), \tag{1.1}$$

and **σ-additive** (a measure on $\mathcal{R}$) if $\varphi(\dot\bigcup_i A_i) = \sum_i \varphi(A_i)$ whenever the $A_i \in \mathcal{R}$ are pairwise disjoint and $\bigcup_i A_i \in \mathcal{R}$.

Remark 1.14

If φ is additive on $\mathcal{R}$, then by induction it is finitely additive, $\varphi(A_1 \dot\cup \cdots \dot\cup A_n) = \sum_i \varphi(A_i)$, and for all $A, B \in \mathcal{R}$:

$$\varphi(A \cup B) + \varphi(A \cap B) = \varphi(A) + \varphi(B), \tag{1.2}$$

$$A \subset B \Rightarrow \varphi(A) \le \varphi(B) \quad \text{(monotonicity)}, \tag{1.3}$$

$$A \subset B,\ \varphi(A) < \infty \Rightarrow \varphi(B \setminus A) = \varphi(B) - \varphi(A),$$

$$\varphi\Big(\bigcup_{i \le n} A_i\Big) \le \sum_{i \le n} \varphi(A_i) \quad \text{(finite subadditivity)}.$$

(1.2): decompose $A \cup B = (A \setminus B) \dot\cup (A \cap B) \dot\cup (B \setminus A)$, $A = (A \setminus B) \dot\cup (A \cap B)$, $B = (B \setminus A) \dot\cup (A \cap B)$. Monotonicity: $\varphi(B) = \varphi(A) + \varphi(B \setminus A)$; subadditivity: (1.2) and induction.

Notation. $A_n \nearrow A$ means $A_n \subset A_{n+1}$ and $A = \bigcup_n A_n$; $A_n \searrow A$ means $A_n \supset A_{n+1}$ and $A = \bigcap_n A_n$.

Theorem 1.15 (Continuity of measures)

Let φ be additive on a ring $\mathcal{R}$ and consider

(i) φ is σ-additive on $\mathcal{R}$;

(ii) (continuity from below) $A_n \in \mathcal{R}$, $A_n \nearrow A \in \mathcal{R} \Rightarrow \varphi(A) = \lim_n \varphi(A_n)$;

(iii) (continuity from above) $A_n \in \mathcal{R}$, $A_n \searrow A \in \mathcal{R}$, $\varphi(A_{n_0}) < \infty$ for some $n_0 \Rightarrow \varphi(A) = \lim_n \varphi(A_n)$.

Then (i) $\Leftrightarrow$ (ii) $\Rightarrow$ (iii).

Proof.

Step (i)⇒(ii). The disjoint $B_1 := A_1$, $B_n := A_n \setminus A_{n-1}$ give $A_n = B_1 \dot\cup \cdots \dot\cup B_n$, $\bigcup_n B_n = A$, so $\lim_n \varphi(A_n) = \lim_n \sum_{i\le n} \varphi(B_i) = \sum_i \varphi(B_i) = \varphi(A)$.

Step (ii)⇒(i). For disjoint $A_n \in \mathcal{R}$ with $A := \bigcup_n A_n \in \mathcal{R}$, $B_N := \bigcup_{n\le N} A_n \nearrow A$, so $\varphi(A) = \lim_N \varphi(B_N) = \lim_N \sum_{n\le N} \varphi(A_n)$.

Step (ii)⇒(iii) [Supplement]. $C_n := A_{n_0} \setminus A_n \nearrow A_{n_0} \setminus A$ $(n \ge n_0)$, all values finite, so by Remark 1.14 and (ii)

$$\varphi(A_{n_0}) - \varphi(A) = \varphi(A_{n_0} \setminus A) = \lim_n \varphi(C_n) = \varphi(A_{n_0}) - \lim_n \varphi(A_n).$$

■

Definition 1.16 (Intervals and elementary sets)

For $a, b \in \mathbb{R}^d$ with $a_i \le b_i$, an **interval** (bounded box) is any set of points x with $a_i \lhd_i x_i \lhd_i' b_i$, $\lhd_i, \lhd_i' \in \{\le, <\}$. It is open (closed) if all inequalities are strict (non-strict); $a_i = b_i$ is allowed, so an interval may be empty or a point. An **elementary set** is a finite union of intervals; $\mathcal{E}$ denotes the family of elementary sets. For an interval I put $\lambda(I) := \prod_{i=1}^d (b_i - a_i)$, and for $A = I_1 \dot\cup \cdots \dot\cup I_n$ with disjoint intervals

$$\lambda(A) := \sum_{i=1}^n \lambda(I_i). \tag{1.4}$$

$\mathcal{E}$ is a ring but not a σ-ring ($\mathbb{Q} \cap [0,1]$). Every $A \in \mathcal{E}$ is a finite *disjoint* union of intervals (cells of the grid of all endpoints), and (1.4) is independent of the decomposition (common grid refinement). So λ is finite, additive and nonnegative on $\mathcal{E}$.

Definition 1.17 (Regularity)

An additive $\varphi : \mathcal{E} \to [0, \infty]$ is **regular** if for every $A \in \mathcal{E}$ and $\varepsilon > 0$ there are $F, G \in \mathcal{E}$, F closed, G open, with $F \subset A \subset G$ and $\varphi(G) - \varepsilon \le \varphi(A) \le \varphi(F) + \varepsilon$.

Example 1.18

λ is regular and finite on $\mathcal{E}$: shrink or enlarge each interval to a closed or open one, changing the volume by less than ε.

From now on μ is a finite, regular, additive set function on $\mathcal{E}$ (e.g. λ).

Definition 1.19 (Outer measure)

For $E \subset \mathbb{R}^d$,

$$\mu^*(E) := \inf\Big\{ \sum_{n=1}^{\infty} \mu(A_n) \;\Big|\; A_n \in \mathcal{E} \text{ open},\ E \subset \bigcup_n A_n \Big\}.$$

Clearly $\mu^*(E) \geq 0$, $\mu^*(\emptyset) = 0$, and $E_1 \subset E_2 \Rightarrow \mu^*(E_1) \leq \mu^*(E_2)$.

Theorem 1.20

(i) $\mu^*(A) = \mu(A)$ for $A \in \mathcal{E}$, so μ^* extends μ to $\mathcal{P}(\mathbb{R}^d)$.

(ii) (σ-subadditivity) If $E = \bigcup_n E_n$ with arbitrary $E_n \subset \mathbb{R}^d$, then

$$\mu^*(E) \leq \sum_{n=1}^{\infty} \mu^*(E_n). \tag{1.5}$$

Proof. (i) By regularity an open $G \supset A$ in $\mathcal{E}$ has $\mu(G) \leq \mu(A) + \varepsilon$, so $\mu^*(A) \leq \mu(A)$.

Conversely, take an open cover $A_n \in \mathcal{E}$ with $\sum_n \mu(A_n) \leq \mu^*(A) + \varepsilon$ and closed $F \subset A$ with $\mu(A) \leq \mu(F) + \varepsilon$. F is compact, so $F \subset A_1 \cup \cdots \cup A_N$ and

$$\mu(A) \leq \mu(F) + \varepsilon \leq \mu(A_1 \cup \cdots \cup A_N) + \varepsilon \leq \sum_{k \leq N} \mu(A_k) + \varepsilon \leq \mu^*(A) + 2\varepsilon.$$

(ii) Let all $\mu^*(E_n) < \infty$ and choose open covers $(A_{nk})_k \subset \mathcal{E}$ of E_n with $\sum_k \mu(A_{nk}) \leq \mu^*(E_n) + 2^{-n}\varepsilon$; together they cover E, so $\mu^*(E) \leq \sum_n \sum_k \mu(A_{nk}) \leq \sum_n \mu^*(E_n) + \varepsilon$. ∎

Definition 1.21 (Approximation by elementary sets)

For $A, B \subset \mathbb{R}^d$ let

$$S(A,B) := (A \setminus B) \cup (B \setminus A) \ \text{ (symmetric difference)}, \qquad d(A,B) := \mu^*(S(A,B)) \in [0,\infty].$$

Write $A_n \to A$ if $d(A, A_n) \to 0$, and define

$$A \in \mathcal{M}_F(\mu) \text{ (finitely } \mu\text{-measurable)} :\Longleftrightarrow \exists A_n \in \mathcal{E} :\ A_n \to A,$$
$$A \in \mathcal{M}(\mu) \text{ } (\mu\text{-measurable)} :\Longleftrightarrow \ A = \textstyle\bigcup_n A_n \text{ with } A_n \in \mathcal{M}_F(\mu).$$

$\mathcal{E} \subset \mathcal{M}_F(\mu) \subset \mathcal{M}(\mu)$, and $A \in \mathcal{M}_F(\mu)$ has $\mu^*(A) < \infty$: $A \subset S(A, A_n) \cup A_n$, so $\mu^*(A) \leq d(A, A_n) + \mu(A_n) < \infty$.

Theorem 1.22

$\mathcal{M}(\mu)$ is a σ-ring and μ^* is σ-additive on $\mathcal{M}(\mu)$.

Lemma 1.23

For $A, B, C \subset \mathbb{R}^d$:

(1) $S(A,B) = S(B,A)$;

(2) $S(A,B) \subset S(A,C) \cup S(C,B)$;

(3) $S(A,B) = S(A^c, B^c)$;

(4) for A_1, A_2, B_1, B_2, each of $S(A_1 \cup A_2, B_1 \cup B_2)$, $S(A_1 \cap A_2, B_1 \cap B_2)$, $S(A_1 \setminus A_2, B_1 \setminus B_2)$

is contained in $S(A_1, B_1) \cup S(A_2, B_2)$.

Consequently, by monotonicity and subadditivity of μ^*: (i) $d(A, B) = d(B, A)$;
(ii) $d(A, B) \le d(A, C) + d(C, B)$;
(iii) $A = B \Rightarrow d(A, B) = 0$ (not conversely);
(iv) $d(A_1 \star A_2, B_1 \star B_2) \le d(A_1, B_1) + d(A_2, B_2)$ for $\star \in \{\cup, \cap, \setminus\}$.

Proof. (1) is clear.

(2) $x \in A \setminus B$ lies in $C \setminus B$ or $A \setminus C$, so $A \setminus B \subset (A \setminus C) \cup (C \setminus B)$; symmetrically $B \setminus A \subset (B \setminus C) \cup (C \setminus A)$.

(3) $S(A, B) = (A \cap B^c) \cup (B \cap A^c) = (B^c \setminus A^c) \cup (A^c \setminus B^c) = S(A^c, B^c)$.

(4) $(A_1 \cup A_2) \setminus (B_1 \cup B_2) \subset (A_1 \setminus B_1) \cup (A_2 \setminus B_2)$ and symmetrically, so

$$S(A_1 \cup A_2, B_1 \cup B_2) \subset S(A_1, B_1) \cup S(A_2, B_2). \tag{1.6}$$

Intersections, by (3) and (1.6): $S(A_1 \cap A_2, B_1 \cap B_2) = S(A_1^c \cup A_2^c, B_1^c \cup B_2^c) \subset S(A_1^c, B_1^c) \cup S(A_2^c, B_2^c) = S(A_1, B_1) \cup S(A_2, B_2)$. Differences: $A_1 \setminus A_2 = A_1 \cap A_2^c$ and (3). ∎

Lemma 1.24

If $\mu^*(A) < \infty$ or $\mu^*(B) < \infty$, then

$$|\mu^*(A) - \mu^*(B)| \le d(A, B). \tag{1.7}$$

Proof. Say $\mu^*(B) \le \mu^*(A)$, so $\mu^*(B) < \infty$; by Lemma 1.23(ii),

$$\mu^*(A) = d(A, \emptyset) \le d(A, B) + d(B, \emptyset) = d(A, B) + \mu^*(B).$$

The other case is symmetric. ∎

Remark 1.25

(i) For $\mu = \lambda$, $d = 1$, a countable $A \subset \mathbb{R}$ and $B = \emptyset$: $d(A, B) = \lambda^*(A) \le \sum_{p \in A} \lambda([p, p]) = 0$, although $A \ne B$: d is only a pseudo-metric.

(ii) On $\mathcal{F} := \{A \subset \mathbb{R}^d \mid \mu^*(A) < \infty\} \supset \mathcal{M}_F(\mu)$, modulo $d(A, B) = 0$, $(\mathcal{F}, d)$ is a metric space and $\mathcal{M}_F(\mu) = \overline{\mathcal{E}}$. [Supplement] *Reason:* d is finite on $\mathcal{F}$ (subadditivity), symmetric with triangle inequality (Lemma 1.23); $\mathcal{M}_F(\mu) \subset \overline{\mathcal{E}}$ by definition, and conversely by a diagonal choice.

Proof of Theorem 1.22.

Step 1 ($\mathcal{M}_F(\mu)$ is a ring). For elementary $A_n \to A$, $B_n \to B$, by Lemmas 1.23(iv) and 1.24,

$$\begin{aligned} &d(A_n \cup B_n, A \cup B) + d(A_n \cap B_n, A \cap B) + d(A_n \setminus B_n, A \setminus B) + |\mu^*(A_n) - \mu^*(A)| + |\mu^*(B_n) - \mu^*(B)| \\ &\quad + |\mu^*(A_n \cup B_n) - \mu^*(A \cup B)| + |\mu^*(A_n \cap B_n) - \mu^*(A \cap B)| \le 6\big(d(A_n, A) + d(B_n, B)\big) \to 0. \end{aligned}$$

Hence $A \cup B$, $A \setminus B \in \mathcal{M}_F(\mu)$ and

$$\mu^*(A_n) \to \mu^*(A), \quad \mu^*(A_n \cup B_n) \to \mu^*(A \cup B), \quad \mu^*(A_n \cap B_n) \to \mu^*(A \cap B). \tag{1.8}$$

Step 2 (μ^* is additive on $\mathcal{M}_F(\mu)$). Pass to the limit in (1.2) for A_n, B_n using (1.8): $\mu^*(A) + \mu^*(B) = \mu^*(A \cup B) + \mu^*(A \cap B)$, and $\mu^*(A \cap B) = 0$ if $A \cap B = \emptyset$.

Step 3 (disjoint representation). Disjointify $A = \bigcup_n A'_n$, $A'_n \in \mathcal{M}_F(\mu)$: $B'_n := A'_1 \cup \cdots \cup A'_n$, $A_1 := B'_1$, $A_n := B'_n \setminus B'_{n-1}$. Then $\mu^*(A) \le \sum_n \mu^*(A_n)$ (σ-subadditivity) and $\sum_{k\le n} \mu^*(A_k) = \mu^*(A_1 \dot\cup \cdots \dot\cup A_n) \le \mu^*(A)$, so

$$\mu^*(A) = \sum_{n=1}^{\infty} \mu^*(A_n). \tag{1.9}$$

Step 4 (finite measure implies $\mathcal{M}_F$). For $\mu^*(A) < \infty$ and $B_n := A_1 \cup \cdots \cup A_n$, $d(A, B_n) = \mu^*(\bigcup_{k>n} A_k) = \sum_{k>n} \mu^*(A_k) \to 0$ by (1.9) (Step 3 applied to $\bigcup_{k>n} A_k$). Elementary B_{n,k_n} with $d(B_{n,k_n}, B_n) < 1/n$ then converge to A:

$$A \in \mathcal{M}(\mu),\ \mu^*(A) < \infty \implies A \in \mathcal{M}_F(\mu). \tag{1.10}$$

Step 5 (σ-additivity on $\mathcal{M}(\mu)$). For disjoint $A_n \in \mathcal{M}(\mu)$ with union A: if $\mu^*(A) = \infty$, $\sum_n \mu^*(A_n) = \infty$ by σ-subadditivity; otherwise $A_n \in \mathcal{M}_F(\mu)$ by (1.10) and (1.9) applies.

Step 6 ($\mathcal{M}(\mu)$ is a σ-ring). $\bigcup_n A_n = \bigcup_{(n,k)\in\mathbb{N}^2} A_{nk} \in \mathcal{M}(\mu)$. For $A = \bigcup_n A_n$, $B = \bigcup_l B_l$ ($A_n, B_l \in \mathcal{M}_F(\mu)$), $A_n \cap B = \bigcup_l (A_n \cap B_l) \in \mathcal{M}(\mu)$ has finite measure, so lies in $\mathcal{M}_F(\mu)$ by (1.10); then $A_n \setminus B = A_n \setminus (A_n \cap B) \in \mathcal{M}_F(\mu)$ and $A \setminus B = \bigcup_n (A_n \setminus B) \in \mathcal{M}(\mu)$. ∎

Definition 1.26 (Lebesgue measure)

$\mu(A) := \mu^*(A)$ on $\mathcal{M}(\mu)$ extends μ to a measure on the σ-ring $\mathcal{M}(\mu)$ (Definition 1.10(ii)); for $\mu = \lambda$ it is the **Lebesgue measure** on $\mathbb{R}^d$.

Remark 1.27

(i) $\mathcal{M}(\mu)$ contains the open sets (countable unions of open intervals, via rational balls), hence $\mathbb{R}^d$: it is a σ-algebra containing the closed sets.

(ii) (*Regularity*) For $A \in \mathcal{M}(\mu)$, $\varepsilon > 0$ there are closed $F \subset A \subset G$ open with $\mu(G \setminus A) < \varepsilon$, $\mu(A \setminus F) < \varepsilon$: G by Proposition 1.28, and $F := \tilde G^c$ for an open $\tilde G \supset A^c$ with $\mu(\tilde G \setminus A^c) < \varepsilon$, since $A \setminus F = \tilde G \setminus A^c$.

(iii) With $G_n \supset A \supset F_n$ for $\varepsilon = 1/n$, $\mu(\bigcap_n G_n \setminus A) = 0 = \mu(A \setminus \bigcup_n F_n)$: $A = B \dot\cup (A \setminus B)$, $B := \bigcup_n F_n \in \mathcal{B}(\mathbb{R}^d)$, $A \setminus B$ null. Measurable = Borel + null, and $\mathcal{B}(\mathbb{R}^d) \subset \mathcal{M}(\mu)$.

(iv) The μ-null sets form a σ-ring, and $\mu^*(N) = 0$ makes N null. [Supplement] *Reason:* σ-subadditivity and monotonicity; $d(N, \emptyset) = 0$ gives $N \in \mathcal{M}_F(\mu)$.

(v) For λ every countable set is a null set, but there are uncountable null sets, e.g. the Cantor set.

Proposition 1.28

μ is regular on $\mathcal{M}(\mu)$: for $A \in \mathcal{M}(\mu)$ and $\varepsilon > 0$ there is an open $G \supset A$ with $\mu(G \setminus A) < \varepsilon$ (and hence Remark 1.27(ii) holds).

Proof.

Step 1 ($\mu(A) < \infty$). $A \in \mathcal{M}_F(\mu)$ by (1.10); open $O_n \in \mathcal{E}$ cover A with $\sum_n \mu(O_n) < \mu(A) + \varepsilon$, so $G := \bigcup_n O_n$ has $\mu(G \setminus A) = \mu(G) - \mu(A) < \varepsilon$.

Step 2 ($\mu(A) = \infty$) [Supplement]. $A = \bigcup_n A_n$, $A_n \in \mathcal{M}_F(\mu)$; Step 1 gives open $G_n \supset A_n$, $\mu(G_n \setminus A_n) < \varepsilon 2^{-n}$, and $G := \bigcup_n G_n$ has $G \setminus A \subset \bigcup_n (G_n \setminus A_n)$. ∎

1.3 Measure spaces and measurable functions

Motivation. On an arbitrary $(X, \mathcal{M}, \mu)$, measurable functions are stable under limits, sums, products and continuous maps.

Example 1.29

(i) $(\mathbb{R}^d, \mathcal{M}(\lambda), \lambda)$ is the **Lebesgue measure space**.
(ii) $\lambda|_{\mathcal{B}(\mathbb{R}^d)}$ is the **Lebesgue–Borel measure**, and $(\mathbb{R}^d, \mathcal{B}(\mathbb{R}^d), \lambda)$ the Lebesgue–Borel measure space (also just "Lebesgue measure").

Definition 1.30 (Measurable function)

An extended real-valued $f : X \to \overline{\mathbb{R}} = [-\infty, \infty]$ is **measurable** if $\{f > \alpha\} := \{x \in X \mid f(x) \in (\alpha, \infty]\} \in \mathcal{M}$ for every $\alpha \in \mathbb{R}$.

For $\mathcal{M}(\lambda)$: *Lebesgue measurable*; for $\mathcal{B}(\mathbb{R}^d)$: *Borel measurable.*

Example 1.31

(i) Borel measurable functions are Lebesgue measurable. The converse fails: $\mathcal{B}(\mathbb{R}^d)$ has the cardinality $\mathfrak{c}$ of the continuum while $\mathcal{M}(\lambda)$ has cardinality $2^{\mathfrak{c}}$ (Dudley, *Real Analysis and Probability*, Prop. 3.4.1), so there is a Lebesgue measurable, non-Borel set L, and $\mathbb{1}_L$ is Lebesgue but not Borel measurable. (ii) A continuous $f : \mathbb{R}^d \to \mathbb{R}$ is Borel measurable, since $\{f > \alpha\} = f^{-1}((\alpha, \infty))$ is open.

Remark 1.32

For measurable spaces $(X, \mathcal{M})$, $(\tilde{X}, \tilde{\mathcal{M}})$, a map $T : X \to \tilde{X}$ is **$\mathcal{M}/\tilde{\mathcal{M}}$-measurable** if $T^{-1}(\tilde{A}) \in \mathcal{M}$ for all $\tilde{A} \in \tilde{\mathcal{M}}$. If $\tilde{\mathcal{M}} = \sigma(\tilde{\mathcal{M}}_0)$, checking $\tilde{A} \in \tilde{\mathcal{M}}_0$ suffices. [Supplement] *Reason:* $\{\tilde{A} \mid T^{-1}(\tilde{A}) \in \mathcal{M}\}$ is a σ-algebra containing $\tilde{\mathcal{M}}_0$. In particular, with $\mathcal{B}(\overline{\mathbb{R}}) := \{A \subset \overline{\mathbb{R}} \mid A \cap \mathbb{R} \in \mathcal{B}(\mathbb{R})\} = \sigma(\{(\alpha, \infty] \mid \alpha \in \mathbb{R}\})$, Definition 1.30 is exactly $\mathcal{M}/\mathcal{B}(\overline{\mathbb{R}})$-measurability.

Theorem 1.33

For $f : X \to \overline{\mathbb{R}}$ the following are equivalent: (i) $\{f > \alpha\} \in \mathcal{M}\ \forall \alpha \in \mathbb{R}$;
(ii) $\{f \geq \alpha\} \in \mathcal{M}\ \forall \alpha$;
(iii) $\{f < \alpha\} \in \mathcal{M}\ \forall \alpha$;
(iv) $\{f \leq \alpha\} \in \mathcal{M}\ \forall \alpha$.

Proof. For $\alpha \in \mathbb{R}$:

(i)⇒(ii): $\{f \geq \alpha\} = \bigcap_n \{f > \alpha - \frac{1}{n}\}$.

(ii)⇒(iii): $\{f < \alpha\} = X \setminus \{f \geq \alpha\}$.

(iii)⇒(iv): $\{f \leq \alpha\} = \bigcap_n \{f < \alpha + \frac{1}{n}\}$.

(iv)⇒(i): $\{f > \alpha\} = X \setminus \{f \leq \alpha\}$. ∎

Theorem 1.34

f measurable $\Rightarrow |f|$ measurable.

Proof. $\{|f| < \alpha\} = \{f < \alpha\} \cap \{f > -\alpha\} \in \mathcal{M}$; apply Theorem 1.33. ■

Theorem 1.35

If f_n, $n \in \mathbb{N}$, are measurable, so are $\inf_n f_n$, $\sup_n f_n$, $\liminf_n f_n$ and $\limsup_n f_n$.

Proof. $\{\sup_n f_n > \alpha\} = \bigcup_n \{f_n > \alpha\}$ and $\{\inf_n f_n < \alpha\} = \bigcup_n \{f_n < \alpha\}$. Then $\limsup_n f_n = \inf_n \sup_{m \geq n} f_m$ and $\liminf_n f_n = \sup_n \inf_{m \geq n} f_m$ are measurable by two applications. ■

Corollary 1.36

(i) f, g measurable $\Rightarrow$ $\max(f, g)$, $\min(f, g)$ measurable; in particular $f^+ := \max(f, 0)$ and $f^- := -\min(f, 0)$ are measurable. (ii) A pointwise limit of measurable functions is measurable.

Proof. (i) is Theorem 1.35 for the sequence $f, g, g, \ldots$.

(ii) $f = \limsup_n f_n$. ■

Theorem 1.37

Let $f, g : X \to \mathbb{R}$ be measurable and $F : \mathbb{R}^2 \to \mathbb{R}$ continuous. Then $h(x) := F(f(x), g(x))$ is measurable; in particular $f + g$ and fg are measurable.

Proof. $G_\alpha := \{F > \alpha\} \subset \mathbb{R}^2$ is open, so (Remark 1.27(i)) a countable union of open rectangles $I_n = (\alpha_n, \beta_n) \times (\gamma_n, \delta_n)$. Each $\{(f, g) \in I_n\} = \{\alpha_n < f < \beta_n\} \cap \{\gamma_n < g < \delta_n\} \in \mathcal{M}$, so $\{h > \alpha\} = \bigcup_n \{(f, g) \in I_n\} \in \mathcal{M}$. ■

Remark 1.38

Compositions of Lebesgue measurable functions need not be Lebesgue measurable. There is a continuous increasing $f : [0, 1] \to [0, 1]$ and a Lebesgue measurable L with $f^{-1}(L)$ not Lebesgue measurable (Dudley, Prop. 4.2.1); then $g = \mathbb{1}_L$ gives $(g \circ f)^{-1}(\{1\}) = f^{-1}(L) \notin \mathcal{M}(\lambda)$. But $g \circ f$ is Lebesgue measurable for Borel g and Lebesgue measurable f (Borel if both are), and every Lebesgue measurable f is a.e. (Definition 1.49) Borel.

Definition 1.39 (Simple functions)

$f : X \to \mathbb{R}$ is **simple** if $\mathrm{im}(f) = \{f(x) \mid x \in X\}$ is finite. If $\mathrm{im}(f) = \{\alpha_1, \ldots, \alpha_n\}$ with distinct α_i and $E_i := \{f = \alpha_i\}$, then

$$f = \sum_{i=1}^{n} \alpha_i \mathbb{1}_{E_i}, \qquad \mathbb{1}_E(x) := \begin{cases} 1 & x \in E \\ 0 & x \notin E \end{cases} \tag{1.11}$$

($\mathbb{1}_E$ is the **indicator** or characteristic function of E). Since $\{\mathbb{1}_E \leq \alpha\}$ is $\emptyset$, E^c or X according as $\alpha < 0$, $0 \leq \alpha < 1$, $\alpha \geq 1$, the function $\mathbb{1}_E$ is measurable iff $E \in \mathcal{M}$; and f in (1.11) is measurable iff $E_1, \ldots, E_n \in \mathcal{M}$.

Theorem 1.40

For every $f : X \to \overline{\mathbb{R}}$ there are simple f_n with $f_n(x) \to f(x)$ for all x. If f is measurable the f_n can be chosen measurable, and if $f \geq 0$ they can be chosen pointwise increasing.

Proof. For $f \geq 0$ let

$$f_n := \sum_{i=1}^{n2^n} \frac{i-1}{2^n} \mathbb{1}_{\{f \in [\frac{i-1}{2^n}, \frac{i}{2^n})\}} + n \mathbb{1}_{\{f \geq n\}}. \tag{1.12}$$

Step 1 (increasing). $n \to n+1$ halves each dyadic cell without lowering values; on $\{f \geq n\}$ the new value is $\geq n$.

Step 2 (convergence). For $n > f(x)$, $0 \leq f(x) - f_n(x) < 2^{-n}$; if $f(x) = \infty$, $f_n(x) = n \to \infty$.

Step 3. The sets are measurable with f; in general take $f_n^+ - f_n^-$. ∎

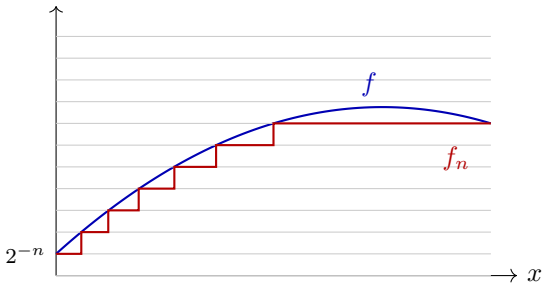


Figure. (1.12): *f_n rounds f down to $2^{-n}\mathbb{Z}$, capped at n.*

Note 1.41

For bounded f, (1.12) converges *uniformly* (error $< 2^{-n}$ once $n > \sup|f|$).

1.4 Integration

Motivation. Approximate from below by simple functions; the payoff is the three limit theorems (MCT, Fatou, DCT).

$(X, \mathcal{M}, \mu)$ is a measure space.

Definition 1.42 (Integral of simple functions)

For $s = \sum_{i=1}^n \alpha_i \mathbb{1}_{E_i}$ with $\alpha_i > 0$, pairwise disjoint $E_i \in \mathcal{M}$, and $E \in \mathcal{M}$, put

$$I_E(s) := \sum_{i=1}^n \alpha_i \mu(E \cap E_i). \tag{1.13}$$

Lemma 1.43

If $s, \tilde{s}$ are measurable simple functions as in Definition 1.42 with $0 \leq s \leq \tilde{s}$, then $I_E(s) \leq I_E(\tilde{s})$. In particular $I_E(s)$ does not depend on the representation of s.

Proof [Supplement]. For $s = \sum_i \alpha_i \mathbb{1}_{E_i}$, $\tilde{s} = \sum_j \beta_j \mathbb{1}_{F_j}$: $E_i \subset \bigcup_j F_j$ and $\alpha_i \leq \beta_j$ on nonempty $E_i \cap F_j$, so

$$I_E(s) = \sum_{i,j} \alpha_i \mu(E \cap E_i \cap F_j) \leq \sum_{i,j} \beta_j \mu(E \cap E_i \cap F_j) \leq \sum_j \beta_j \mu(E \cap F_j) = I_E(\tilde{s}).$$

Two representations: apply both ways. ■

For measurable $f \geq 0$ define

$$\int_E f\,d\mu := \sup\{I_E(s) \mid s \text{ simple, measurable, } 0 \leq s \leq f\}. \tag{1.14}$$

Remark 1.44

$\int_E f\,d\mu \in [0,\infty]$, and $I_E(s) = I_E(s\mathbb{1}_E)$, so

$$\int_E f\,d\mu = \sup\{I_E(s) \mid s \text{ simple, measurable, } 0 \leq s \leq f\mathbb{1}_E\}. \tag{1.15}$$

Hence for measurable $g \geq 0$:

$$f = g \text{ on } E \Rightarrow \int_E f\,d\mu = \int_E g\,d\mu, \qquad f \leq g \text{ on } E \Rightarrow \int_E f\,d\mu \leq \int_E g\,d\mu, \tag{1.16}$$

and by Lemma 1.43, $\int_E s\,d\mu = I_E(s)$ for every measurable simple $s \geq 0$.

Definition 1.45 (Integral and integrability)

For measurable f with $\min\left(\int_E f^+ d\mu, \int_E f^- d\mu\right) < \infty$ put

$$\int_E f\,d\mu := \int_E f^+ d\mu - \int_E f^- d\mu \in [-\infty,\infty]. \tag{1.17}$$

If both integrals are finite, f is **(Lebesgue) integrable** on E w.r.t. μ, written $f \in \mathcal{L}(E,\mu)$ ($f \in \mathcal{L}(E)$ if $\mu = \lambda$).

Remark 1.46

For $E \in \mathcal{M}$:

(i) f measurable and bounded on E, $\mu(E) < \infty \Rightarrow f \in \mathcal{L}(E,\mu)$.
(ii) f measurable, $\alpha \leq f \leq \beta$ on E, $\mu(E) < \infty \Rightarrow \alpha\mu(E) \leq \int_E f\,d\mu \leq \beta\mu(E)$.
(iii) $f, g \in \mathcal{L}(E,\mu)$, $f \leq g$ on $E \Rightarrow \int_E f\,d\mu \leq \int_E g\,d\mu$.
(iv) $f \in \mathcal{L}(E,\mu) \Rightarrow \alpha f \in \mathcal{L}(E,\mu)$ and $\int_E \alpha f\,d\mu = \alpha \int_E f\,d\mu$ for $\alpha \in \mathbb{R}$.
(v) $\mu(E) = 0$, f measurable $\Rightarrow \int_E f\,d\mu = 0$.
(vi) $f \in \mathcal{L}(E,\mu)$, $A \subset E$, $A \in \mathcal{M} \Rightarrow f \in \mathcal{L}(A,\mu)$.

[Supplement] *Proofs.*
(ii) $0 \leq s \leq c\mathbb{1}_E$ gives $I_E(s) \leq c\mu(E)$, attained by $c\mathbb{1}_E$: $\int_E c\,d\mu = c\mu(E)$. With $f^+ \leq \beta^+$, $f^- \geq \beta^-$, $f^+ \geq \alpha^+$, $f^- \leq \alpha^-$ on E and (1.16), $\alpha\mu(E) \leq \int_E f \leq \beta^+\mu(E) - \beta^-\mu(E) = \beta\mu(E)$.
(i) follows from the bounds in (ii).
(iii) $f^+ \leq g^+$ and $f^- \geq g^-$; use (1.16).
(iv) For $\alpha > 0$, $s \mapsto \alpha s$ matches simple functions below h and αh, $I_E(\alpha s) = \alpha I_E(s)$, and $(\alpha f)^\pm = \alpha f^\pm$; for $\alpha < 0$, $(\alpha f)^\pm = |\alpha| f^\mp$.
(v) Every $I_E(s) = \sum \alpha_i \mu(E \cap E_i) = 0$.
(vi) $I_A(s) \leq I_E(s)$ for $A \subset E$, so $\int_A f^\pm \leq \int_E f^\pm < \infty$.

Theorem 1.47

(i) If $f \geq 0$ is measurable, then $\varphi(A) := \int_A f\,d\mu$, $A \in \mathcal{M}$, is σ-additive, i.e. a measure on $(X, \mathcal{M})$. (ii) The same holds for $f \in \mathcal{L}(X, \mu)$, except that φ is then a difference of two finite measures.

Proof. (ii) is (i) for $f^{\pm}$. For (i) let $A = \dot{\bigcup}_i A_i$.

Step 1 (simple f). $f = \mathbb{1}_B$: $\varphi(A) = \mu(A \cap B) = \sum_i \mu(A_i \cap B)$; sum over $f = \sum_j \beta_j \mathbb{1}_{B_j}$, $\beta_j \geq 0$ (nonnegative series rearrange).

Step 2 ($\leq$). For simple $0 \leq s \leq f$, $\int_A s = \sum_i \int_{A_i} s \leq \sum_i \varphi(A_i)$; take $\sup_s$.

Step 3 ($\geq$). φ is monotone, so assume all $\varphi(A_n) < \infty$. Choose simple $0 \leq s_k \leq f$ with $\int_{A_k} s_k \geq \varphi(A_k) - \varepsilon$; $s := \mathbb{1}_{A_1} s_1 + \mathbb{1}_{A_2} s_2 \leq f$ and by Step 1

$$\varphi(A_1 \cup A_2) \geq \int_{A_1 \cup A_2} s\,d\mu = \int_{A_1} s_1\,d\mu + \int_{A_2} s_2\,d\mu \geq \varphi(A_1) + \varphi(A_2) - 2\varepsilon.$$

Induction and monotonicity: $\varphi(A) \geq \sum_{i \leq n} \varphi(A_i)$ for all n. ∎

Corollary 1.48

Let $A, E \in \mathcal{M}$, $A \subset E$, $\mu(E \setminus A) = 0$, and let $\int_E f\,d\mu$ be defined (e.g. $f \geq 0$ measurable or $f \in \mathcal{L}(E, \mu)$). Then $\int_E f\,d\mu = \int_A f\,d\mu$.

Proof. For $g = f^{\pm}$: $\int_E g = \int_A g + \int_{E \setminus A} g = \int_A g$ by Theorem 1.47 and Remark 1.46(v). ∎

Definition 1.49 (Almost everywhere)

A property P of points of $E \in \mathcal{M}$ holds **μ-almost everywhere** (μ-a.e.) on E if there is $N \in \mathcal{M}$ with $\mu(N) = 0$ such that P holds on $E \setminus N$ ("a.e." for Lebesgue measure).

Example 1.50

For measurable f, g: $f = g$ μ-a.e. on $E \iff \mu(\{f \neq g\} \cap E) = 0$. Here $\{f \neq g\} \in \mathcal{M}$ because $\{f < g\} = \bigcup_{q \in \mathbb{Q}} \{f < q\} \cap \{q < g\}$ and similarly for $\{f > g\}$. [Supplement]

Remark 1.51

"$f \sim g$ on $E :\iff f = g$ μ-a.e. on E" is an equivalence relation (unions of null sets are null). If $f = g$ off a null N and $\int_E f$ is defined, Corollary 1.48 and (1.16) give

$$\int_E f\,d\mu = \int_{E \setminus N} f\,d\mu = \int_{E \setminus N} g\,d\mu = \int_E g\,d\mu.$$

Theorem 1.52

Let $E \in \mathcal{M}$.

(i) If f is measurable, $g \in \mathcal{L}(E, \mu)$ and $|f| \leq g$ μ-a.e. on E, then $f \in \mathcal{L}(E, \mu)$.

(ii) $|f| \in \mathcal{L}(E, \mu)$ (i.e. $\int_E |f|\,d\mu < \infty$) $\iff$ $f \in \mathcal{L}(E, \mu)$.

(iii) $f \in \mathcal{L}(E, \mu) \Rightarrow |f| < \infty$ μ-a.e. on E.

(iv) $f \in \mathcal{L}(E, \mu) \Rightarrow \left|\int_E f\,d\mu\right| \leq \int_E |f|\,d\mu$.

Proof. (i) With $|f| \le g$ off a null N: $\int_E |f| = \int_{E\setminus N} |f| \le \int_{E\setminus N} g = \int_E g < \infty$. Since $f^{\pm} \le |f|$, (1.16) gives $f^{\pm} \in \mathcal{L}(E,\mu)$.

(ii) "$\Rightarrow$" by (i). "$\Leftarrow$": with $A := E \cap \{f \ge 0\}$, $B := E \cap \{f < 0\}$, $\int_E |f| = \int_A f^+ + \int_B f^- < \infty$ by Theorem 1.47 and Remark 1.46(vi).

(iii) For every n, $0 \le n\mathbb{1}_{\{|f|=\infty\}} \le |f|$, so $n\,\mu(\{|f| = \infty\} \cap E) \le \int_E |f| < \infty$; let $n \to \infty$.

(iv) By Remark 1.46(iii),(iv): $\pm \int_E f = \int_E (\pm f) \le \int_E |f|$. ∎

Theorem 1.53 (Monotone convergence theorem, Beppo Levi)

Let $0 \le f_1 \le f_2 \le \dots$ be measurable and $f := \lim_n f_n$. Then $\int_E f\,d\mu = \lim_n \int_E f_n\,d\mu$ for all $E \in \mathcal{M}$.

Proof. $\int_E f_n \nearrow \alpha \le \int_E f$. Conversely, for simple $0 \le s \le f$, $c \in (0,1)$, $E_n := E \cap \{f_n \ge cs\} \nearrow E$ (if $s(x) > 0$, $f_n(x) \to f(x) > cs(x)$), and

$$\int_E f_n\,d\mu \ge \int_{E_n} f_n\,d\mu \ge c \int_{E_n} s\,d\mu.$$

Continuity from below for the measure $A \mapsto \int_A s$ (Theorem 1.47) gives $\alpha \ge c \int_E s$; let $c \uparrow 1$, take $\sup_s$. ∎

Theorem 1.54 (Linearity)

Let $f_1, f_2 \in \mathcal{L}(E,\mu)$ with $f_1 + f_2$ well defined (no $\infty - \infty$). Then:

(i) $f_1 + f_2$ is measurable;

(ii) if f is measurable and $f = f_1 + f_2$ μ-a.e. on E, then $f \in \mathcal{L}(E,\mu)$ and

$$\int_E f\,d\mu = \int_E f_1\,d\mu + \int_E f_2\,d\mu. \tag{1.18}$$

(1.18) also holds if $f_1, f_2 \ge 0$ are merely measurable and f is measurable with $f = f_1 + f_2$ μ-a.e. on E.

Remark 1.55

Measurability of f in Theorem 1.54(ii) is automatic if $(X, \mathcal{M}, \mu)$ is **complete** (subsets of null sets are measurable): then $h = g$ a.e. with g measurable makes h measurable. [Supplement] *Reason:* if $h = g$ off a null N, $\{h > \alpha\} = (\{g > \alpha\} \setminus N) \cup (\{h > \alpha\} \cap N)$, the second set lying in N.

Proof of Theorem 1.54. (i) $f_1^+ - f_1^- + f_2^+ - f_2^-$ is a pointwise limit of simple combinations (Theorem 1.40; Corollary 1.36(ii)).

(ii) By Remark 1.51 take $f = f_1 + f_2$.

Step 1 ($f_1, f_2 \ge 0$). Simple: (1.13). In general simple $s_n^i \nearrow f_i$ and the MCT give $\int_E (f_1 + f_2) = \lim_n \int_E (s_n^1 + s_n^2) = \int_E f_1 + \int_E f_2$.

Step 2 (general). Take f_1, f_2 finite (Theorem 1.52(iii), Remark 1.51). $|f_1 + f_2| \le |f_1| + |f_2|$ gives $f \in \mathcal{L}(E,\mu)$; integrate $f^+ + f_1^- + f_2^- = f_1^+ + f_2^+ + f^-$ by Step 1 and rearrange the finite terms. ∎

Theorem 1.56

For measurable $f_k \geq 0$ and $E \in \mathcal{M}$: $\int_E \sum_{k=1}^{\infty} f_k \, d\mu = \sum_{k=1}^{\infty} \int_E f_k \, d\mu$.

Proof. $\sum_{k \leq n} f_k \nearrow \sum_k f_k$; apply the MCT and Theorem 1.54 to the partial sums. ∎

Theorem 1.57 (Fatou's lemma)

For measurable $f_n \geq 0$ and $E \in \mathcal{M}$: $\int_E \liminf_n f_n \, d\mu \leq \liminf_n \int_E f_n \, d\mu$.

Proof. $g_n := \inf_{k \geq n} f_k$ is measurable, $g_n \nearrow \liminf_n f_n$, and $g_n \leq f_k$ for $k \geq n$. By the MCT and (1.16),

$$\int_E \liminf_n f_n \, d\mu = \lim_n \int_E g_n \, d\mu \leq \liminf_n \int_E f_k \, d\mu = \liminf_n \int_E f_n \, d\mu.$$

∎

Theorem 1.58 (Dominated convergence theorem)

Let f_n be measurable with $f_n(x) \to f(x)$ for all $x \in E$, and suppose there is $g \in \mathcal{L}(E, \mu)$ (an *integrable majorant*) with

$$|f_n(x)| \leq g(x) \qquad \forall n \geq 1, \ x \in E. \tag{1.19}$$

Then $f \in \mathcal{L}(E, \mu)$ and $\int_E f \, d\mu = \lim_n \int_E f_n \, d\mu$.

Remark 1.59

In the MCT and DCT, a.e. convergence suffices if the limit is measurable (automatic when complete, Remark 1.55); (1.19) and $f_n \geq 0$ in Fatou need only hold a.e. [Supplement] *Reason:* multiplying by $\mathbb{1}_{N^c}$, N the union of the exceptional null sets, changes no integral (Corollary 1.48).

Proof of Theorem 1.58. $f, f_n \in \mathcal{L}(E, \mu)$ by Theorem 1.52(i), since $|f| \leq g$ as well.

Step 1. $f_n + g \geq 0$ and $f + g = \liminf_n (f_n + g)$. Fatou and Theorem 1.54 give $\int_E f + \int_E g \leq \liminf_n \int_E f_n + \int_E g$, i.e. $\int_E f \leq \liminf_n \int_E f_n$.

Step 2. $g - f_n \geq 0$ and $g - f = \liminf_n (g - f_n)$, so $\int_E g - \int_E f \leq \int_E g + \liminf_n \int_E (-f_n)$, i.e. $\int_E f \geq \limsup_n \int_E f_n$.

Step 3. Hence $\liminf_n \int_E f_n = \limsup_n \int_E f_n = \int_E f$. ∎

1.4.1 Comparison of the Lebesgue and Riemann integrals

Motivation. Lebesgue agrees with Riemann, and Riemann integrability means a null discontinuity set.

$X = [a, b]$, $\mu = \lambda$ on $\mathcal{M} = \{A \cap [a, b] \mid A \in \mathcal{M}(\lambda)\}$ (complete); $\int_a^b f \, dx := \int_X f \, d\lambda$, $f \in \mathcal{R}([a, b])$ if f is Riemann integrable, $\mathcal{R}\int_a^b f \, dx$ for the Riemann integral and $\underline{\mathcal{R}}\int$, $\overline{\mathcal{R}}\int$ for the lower and upper Riemann integrals. For a partition $P = \{x_0, \ldots, x_n\}$, $\mathrm{mesh}(P) := \max_i |x_i - x_{i-1}|$.

Theorem 1.60

(i) $f \in \mathcal{R}([a,b]) \Rightarrow f \in \mathcal{L}([a,b])$ and $\int_a^b f\,dx = \mathcal{R}\int_a^b f\,dx$.

(ii) For bounded f: $f \in \mathcal{R}([a,b]) \iff f$ is continuous a.e. on $[a,b]$.

Proof.

Step 1 (step functions). Let f be bounded. Choose partitions $P_k \subset P_{k+1}$ with $\operatorname{mesh}(P_k) \le 1/k$ such that the lower and upper sums satisfy

$$L(P_k,f) \to \underline{\mathcal{R}}\int_a^b f\,dx, \qquad U(P_k,f) \to \overline{\mathcal{R}}\int_a^b f\,dx. \tag{1.20}$$

For $P_k = \{x_0,\dots,x_{n(k)}\}$ define step functions L_k, U_k by $L_k(a) = U_k(a) = f(a)$ and, on $(x_{i-1},x_i]$, $L_k := m_i = \inf_{[x_{i-1},x_i]} f$, $U_k := M_i = \sup_{[x_{i-1},x_i]} f$. Then

$$L(P_k,f) = \int_a^b L_k\,dx, \qquad U(P_k,f) = \int_a^b U_k\,dx, \qquad L_1 \le L_2 \le \cdots \le f \le \cdots \le U_2 \le U_1. \tag{1.21}$$

Step 2 (limits). $L := \lim_k L_k$, $U := \lim_k U_k$ are measurable; with the bounded majorant $\max(|L_1|,|U_1|)$, the DCT, (1.21), (1.20) give

$$\int_a^b L\,dx = \underline{\mathcal{R}}\int_a^b f\,dx, \qquad \int_a^b U\,dx = \overline{\mathcal{R}}\int_a^b f\,dx. \tag{1.22}$$

Step 3 (i). Since $L \le f \le U$,

$$f \in \mathcal{R}([a,b]) \iff \int_a^b L\,dx = \int_a^b U\,dx \iff L = U \text{ a.e.} \iff L = f = U \text{ a.e.}, \tag{1.23}$$

the middle step by Lemma 1.62(i) for $U - L \ge 0$. Then f is measurable (completeness, Remark 1.55) and $\int_a^b f\,dx = \int_a^b L\,dx = \mathcal{R}\int_a^b f\,dx$ by Remark 1.51 and (1.22).

Step 4 (ii). Off the countable (hence null) set $\bigcup_k P_k$,

$$U(x) = L(x) \iff f \text{ is continuous at } x. \tag{1.24}$$

[Supplement] Such x is interior to an interval of each P_k, of length $\le 1/k$. Continuity at x puts it, for large k, where $|f - f(x)| < \varepsilon$, so $U_k(x) - L_k(x) \le 2\varepsilon$; conversely $U_k(x) - L_k(x) < \varepsilon$ gives $|f(y) - f(x)| \le M_i - m_i < \varepsilon$ around x. With (1.23): $f \in \mathcal{R}([a,b]) \iff L = U$ a.e. $\iff f$ continuous a.e. ∎

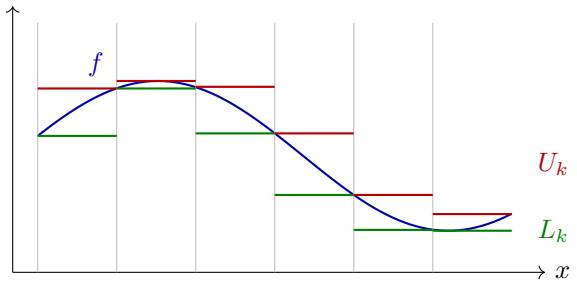


Figure. *Lower and upper step functions; refinement squeezes them to f at continuity points.*

1.5 Some fundamental inequalities and the L^p-spaces

Motivation. Markov, Jensen, Hölder and Minkowski control $\int |f|^p$, and L^p is complete.

Proposition 1.61 (Markov ($p=1$), Chebyshev ($p=2$))

For measurable f, $p \in (0,\infty)$ and $\varepsilon > 0$:

$$\mu(\{|f| \geq \varepsilon\}) \leq \frac{1}{\varepsilon^p} \int_X |f|^p \, d\mu.$$

Proof. $\mathbb{1}_{\{|f|\geq\varepsilon\}} = \mathbb{1}_{\{|f|^p\geq\varepsilon^p\}} \leq \varepsilon^{-p}|f|^p$. Integrate ((1.16); Remark 1.46(iv) for $h \geq 0$ needs no integrability). ■

Lemma 1.62

Let f be measurable. (i) If $f \geq 0$ μ-a.e. and $\int_X f \, d\mu = 0$, then $f = 0$ μ-a.e. (ii) If $\int_A f \, d\mu = 0$ for every $A \in \mathcal{M}$, then $f = 0$ μ-a.e.

Proof [Supplement]. (i) $|f| = f$ a.e., so $\int |f| \, d\mu = 0$ by Remark 1.51. Markov gives $\mu(\{|f| \geq \frac{1}{n}\}) \leq n \int |f| = 0$, and $\{f \neq 0\} = \bigcup_n \{|f| \geq \frac{1}{n}\}$.

(ii) $A := \{f > 0\}$: $\int f\mathbb{1}_A = 0$, so $f\mathbb{1}_A = 0$ a.e. by (i), and $f\mathbb{1}_A > 0$ on A forces $\mu(A) = 0$; likewise $\{f < 0\}$. ■

Theorem 1.63 (Jensen's inequality)

Let $(X, \mathcal{M}, \mu)$ be a probability space, $f \in \mathcal{L}(X,\mu)$ real-valued with $f(x) \in (a,b)$ for all x, and $\varphi : (a,b) \to \mathbb{R}$ convex. Then $(\varphi \circ f)^- \in \mathcal{L}(X,\mu)$ and

$$\varphi\Big(\int_X f \, d\mu\Big) \leq \int_X \varphi \circ f \, d\mu \in (-\infty, \infty].$$

Proof.

Step 1. φ is continuous, so $\varphi \circ f$ is measurable. $t := \int_X f \, d\mu \in (a,b)$, and with the right derivative β_+ at t, $\varphi(y) \geq \varphi(t) + \beta_+(y-t)$, so

$$\varphi \circ f \geq g := \varphi(t) + \beta_+(f - t) \in \mathcal{L}(X,\mu). \tag{1.25}$$

Step 2. $(\varphi \circ f)^- \leq g^-$, so $\int \varphi \circ f \in (-\infty, \infty]$ is defined.

Step 3. g finite a.e., so $(\varphi\circ f)^+ + g^- \geq (\varphi\circ f)^- + g^+ \geq 0$ a.e.; integrate (Theorem 1.54, nonnegative case) and rearrange the finite terms:

$$\varphi\Big(\int_X f \, d\mu\Big) = \int_X g \, d\mu \leq \int_X \varphi \circ f \, d\mu.$$

■

Example 1.64 (AM–GM)

Let $X = \{p_1, \ldots, p_n\}$, $\mu(\{p_i\}) = \alpha_i > 0$ with $\sum \alpha_i = 1$, $f(p_i) = x_i$, $\varphi = \exp$. Jensen gives $e^{\sum \alpha_i x_i} \leq \sum \alpha_i e^{x_i}$, i.e. with $y_i = e^{x_i}$,

$$\prod_{i=1}^n y_i^{\alpha_i} \leq \sum_{i=1}^n \alpha_i y_i \qquad (\alpha_i, y_i > 0,\ \textstyle\sum \alpha_i = 1).$$

Theorem 1.65 (Hölder and Minkowski inequalities)

Let $f, g : X \to [0,\infty]$ be measurable and $p, q \in (1,\infty)$ with $\frac{1}{p} + \frac{1}{q} = 1$. Then

$$\int_X fg\,d\mu \le \Big(\int_X f^p\,d\mu\Big)^{1/p}\Big(\int_X g^q\,d\mu\Big)^{1/q}, \tag{1.26}$$

$$\Big(\int_X (f+g)^p\,d\mu\Big)^{1/p} \le \Big(\int_X f^p\,d\mu\Big)^{1/p} + \Big(\int_X g^p\,d\mu\Big)^{1/p}. \tag{1.27}$$

Proof.

Step 1 (Hölder, trivial cases). A, B: the right factors of (1.26). $A = 0$ gives $f = 0$ a.e. (Lemma 1.62), so $fg = 0$ a.e.; same for B; A or $B = \infty$ is trivial. Else $F := f/A$, $G := g/B$ have $\int F^p = \int G^q = 1$.

Step 2 (Young). $F = e^{s/p}$, $G = e^{t/q}$: $FG \le \frac{1}{p}e^s + \frac{1}{q}e^t = \frac{1}{p}F^p + \frac{1}{q}G^q$ (also for $F, G \in \{0,\infty\}$); integrate: $\int FG \le 1$.

Step 3 (Minkowski). $(f+g)^p = f(f+g)^{p-1} + g(f+g)^{p-1}$, (1.26) and $(p-1)q = p$:

$$\int (f+g)^p \le \Big[\Big(\int f^p\Big)^{1/p} + \Big(\int g^p\Big)^{1/p}\Big]\Big(\int (f+g)^p\Big)^{1/q}. \tag{1.28}$$

Step 4. Assume $\int (f+g)^p > 0$, $\int f^p + \int g^p < \infty$; $(\frac{f+g}{2})^p \le \frac{1}{2}(f^p + g^p)$ gives $C := (\int (f+g)^p)^{1/q} \in (0,\infty)$; divide (1.28) by C. ∎

Definition 1.66 ($\mathcal{L}^p$)

For $p \in [1,\infty)$ let $\mathcal{L}^p(X,\mu) := \{f : X \to \mathbb{R} \text{ measurable} \mid \|f\|_p := (\int_X |f|^p\,d\mu)^{1/p} < \infty\}$.

Theorem 1.67

(i) For $p \in [1,\infty)$, $\mathcal{L}^p(X,\mu)$ is a vector space and $\|\cdot\|_p$ a semi-norm on it. If (f_n) is Cauchy in $\mathcal{L}^p$, there are $f \in \mathcal{L}^p(X,\mu)$ and a subsequence (f_{n_k}) with $\|f - f_n\|_p \to 0$ and $f_{n_k} \to f$ μ-a.e.

(ii) If $p, q \in (1,\infty)$, $\frac{1}{p} + \frac{1}{q} = 1$, $f \in \mathcal{L}^p$, $g \in \mathcal{L}^q$, then $fg \in \mathcal{L}^1$ and

$$\|fg\|_1 \le \|f\|_p \|g\|_q. \tag{1.29}$$

Proof.

Step 1 (semi-norm). $\|\alpha f\|_p = |\alpha|\|f\|_p$; $|f+g| \le |f| + |g|$ and Minkowski give the triangle inequality ($p = 1$: linearity). $\|f\|_p = 0$ gives only $f = 0$ a.e. (Lemma 1.62(i)).

Step 2 (a.e. convergent subsequence). $g_k := f_{n_{k+1}} - f_{n_k}$ with $\|g_k\|_p < 2^{-k}$; by the MCT and Minkowski,

$$\Big\|\sum_k |g_k|\Big\|_p = \lim_n \Big\|\sum_{k\le n} |g_k|\Big\|_p \le \limsup_n \sum_{k\le n} \|g_k\|_p < 1.$$

so (Theorem 1.52(iii)) off a null N, $f_{n_{k+1}} = f_{n_1} + \sum_{l\le k} g_l$ converges; $f := \lim_k f_{n_k} \mathbb{1}_{X\setminus N}$.

Step 3 (L^p convergence). If $\|f_m - f_n\|_p < \varepsilon$ for $m, n \ge N_0$ and $n_k \ge N_0$, Fatou gives $\|f - f_{n_k}\|_p \le \liminf_l \|f_{n_l} - f_{n_k}\|_p \le \varepsilon$. So $f \in \mathcal{L}^p$, $f_{n_k} \to f$, and the Cauchy sequence $f_n \to f$.

(ii) is (1.26) applied to $|f|, |g|$. ∎

Remark 1.68

(i) Identifying functions that agree μ-a.e. turns $\mathcal{L}^p$ into a normed space: $L^p(X,\mu) := \mathcal{L}^p(X,\mu)/\|\cdot\|_p^{-1}(0)$, where $\|\cdot\|_p^{-1}(0) = \{f \text{ measurable} \mid f = 0\ \mu\text{-a.e.}\}$. For the class $\tilde{f}$ of f put

$$\|\tilde{f}\|_p := \|f\|_p; \tag{1.30}$$

now $\|\tilde{f}\|_p = 0 \iff f = 0$ a.e. $\iff \tilde{f} = 0$. In particular $L^2(X,\mu)$ is an inner product space with

$$\langle \tilde{f}, \tilde{g}\rangle_2 := \int_X f g \, d\mu, \qquad \|\tilde{f}\|_2 = \sqrt{\langle \tilde{f}, \tilde{f}\rangle_2}, \tag{1.31}$$

for arbitrary representatives f, g.

(ii) We work with representatives modulo $f = g$ μ-a.e.

(iii) $\mathcal{L}^p$ is real-valued so sums are defined; $g : X \to [0,\infty]$ with $\int g^p < \infty$ is finite a.e. (Theorem 1.52(iii)) and tacitly replaced by a finite version.

Example 1.69

For $X = \mathbb{N}$, $\mathcal{M} = \mathcal{P}(\mathbb{N})$, $\mu = \sum_k \delta_k$ (counting measure), $\emptyset$ is the only null set and

$$L^p(X,\mu) = \mathcal{L}^p(X,\mu) = \ell^p := \Big\{ x = (x_k) \in \mathbb{R}^{\mathbb{N}} \;\Big|\; \sum_k |x_k|^p < \infty \Big\}.$$

Theorem 1.70 (Riesz–Fischer)

For $p \in [1,\infty)$, $(L^p(X,\mu), \|\cdot\|_p)$ is a Banach space; in particular $L^2(X,\mu)$ is a Hilbert space with the inner product (1.31).

Proof. Normed by Remark 1.68(i). Cauchy $(\tilde{f}_n)$ have Cauchy representatives ((1.30)), which converge by Theorem 1.67(i). ∎